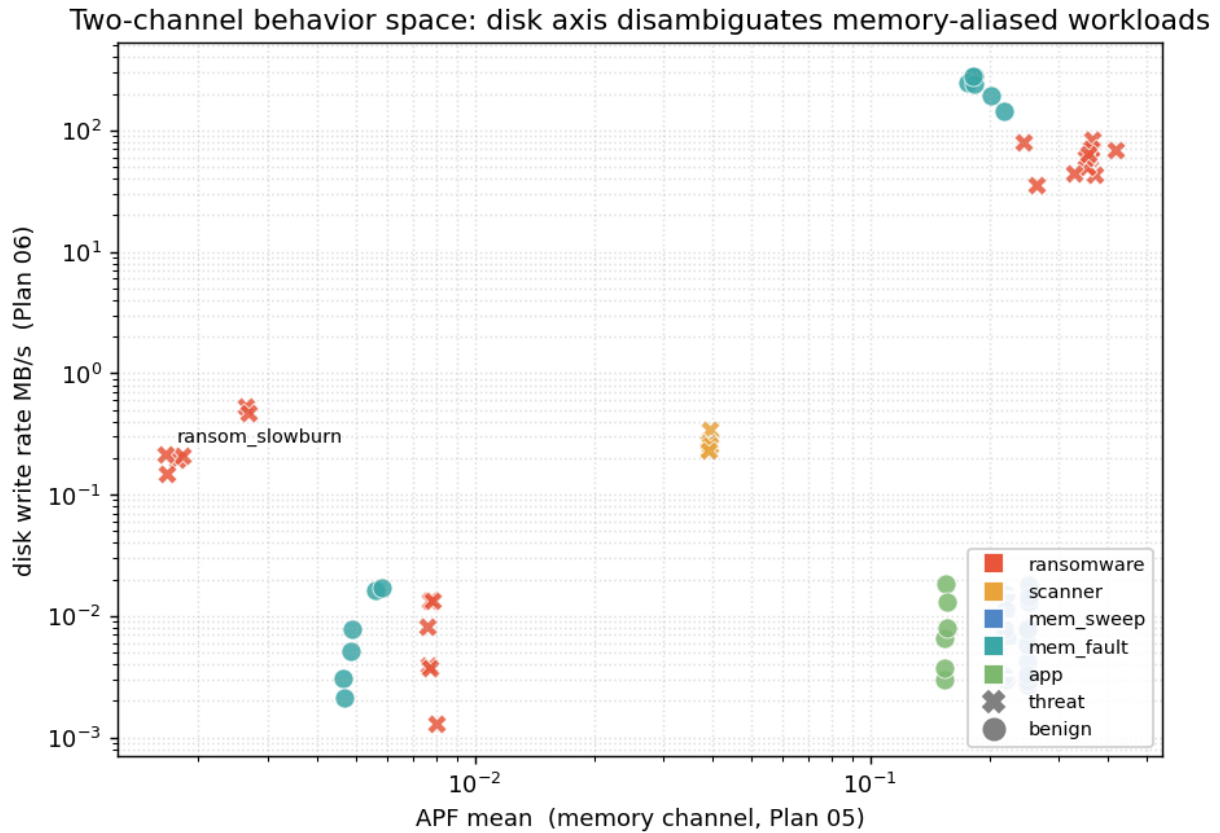


Plan 05 → Plan 06 — Comprehensive Data

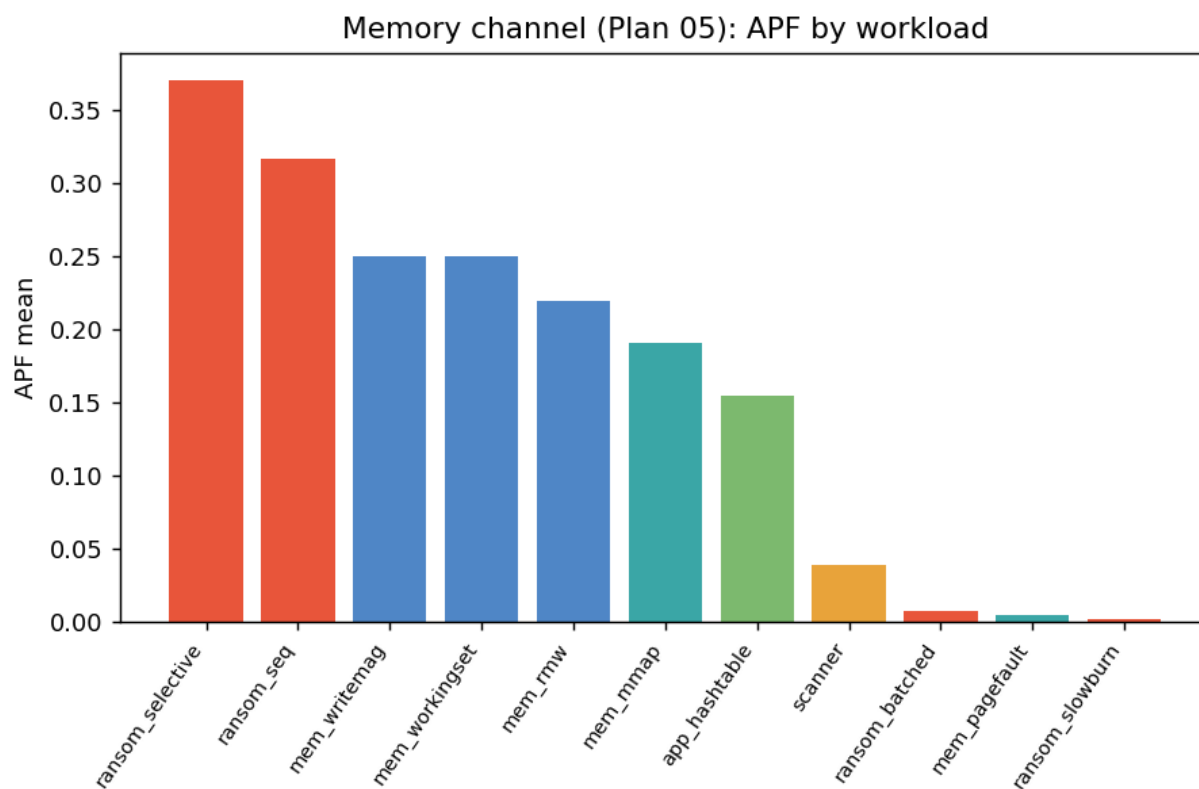
All 66 cells, 11 workloads, both channels. Memory (APF) = Plan 05; disk = Plan 06.

1. Two-channel behavior space

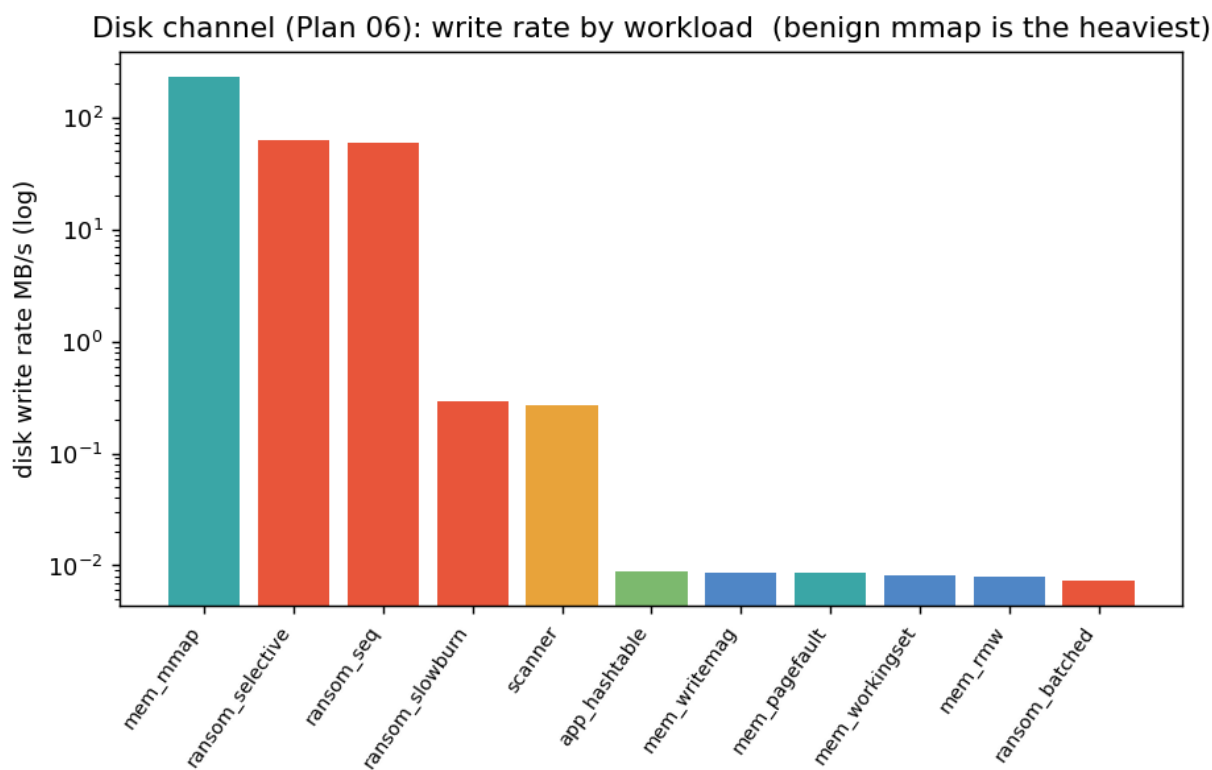


Each point = one cell. X = APF mean (memory), Y = disk write rate (Plan 06). Memory-aliased workloads (slowburn vs pagefault) separate on the disk axis; the heaviest writer is benign (mmap).

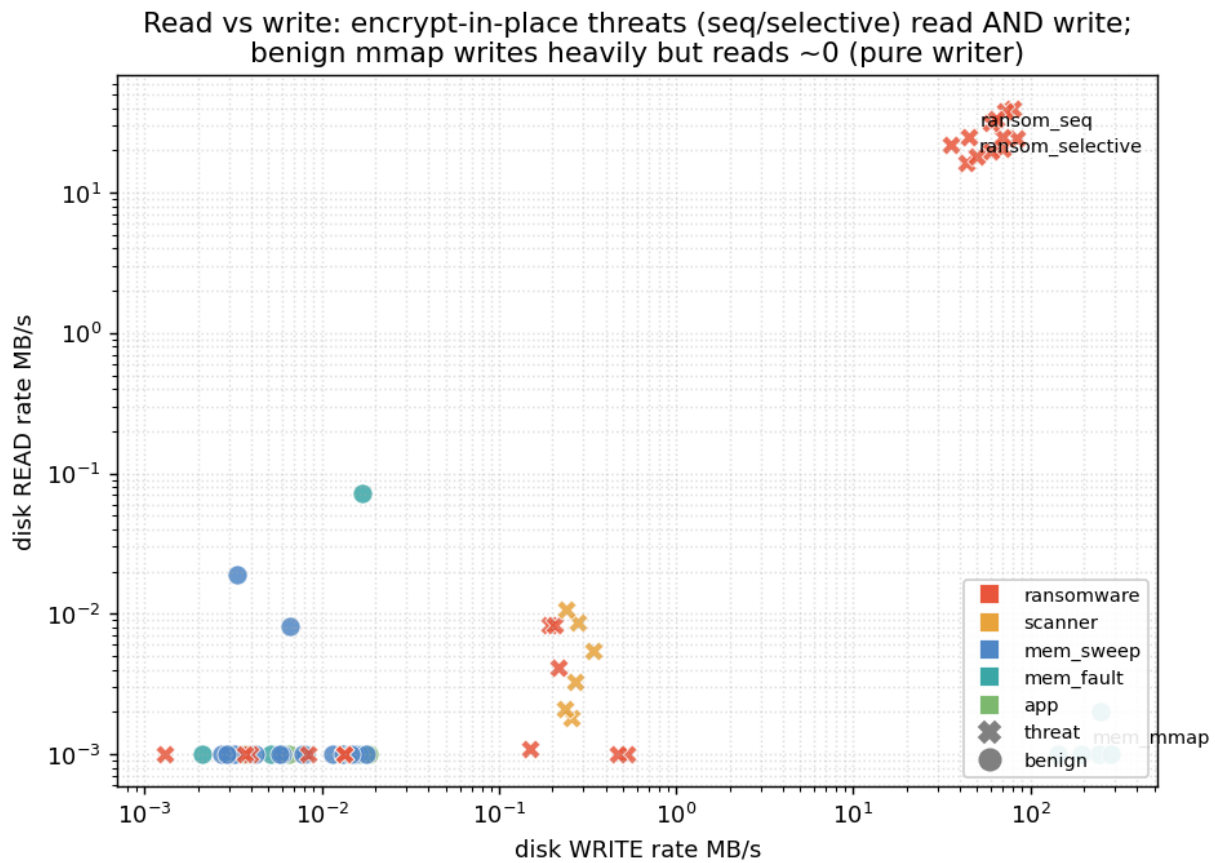
2. Per-workload, both channels



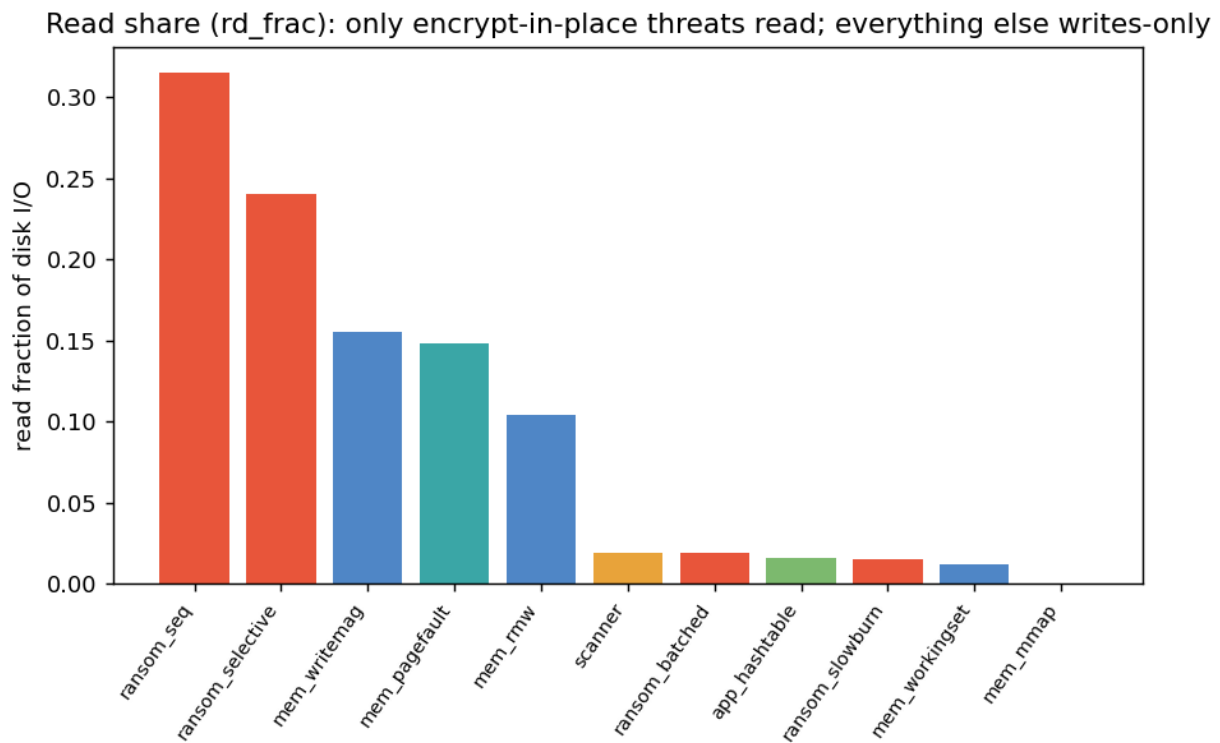
Memory: APF mean per workload.



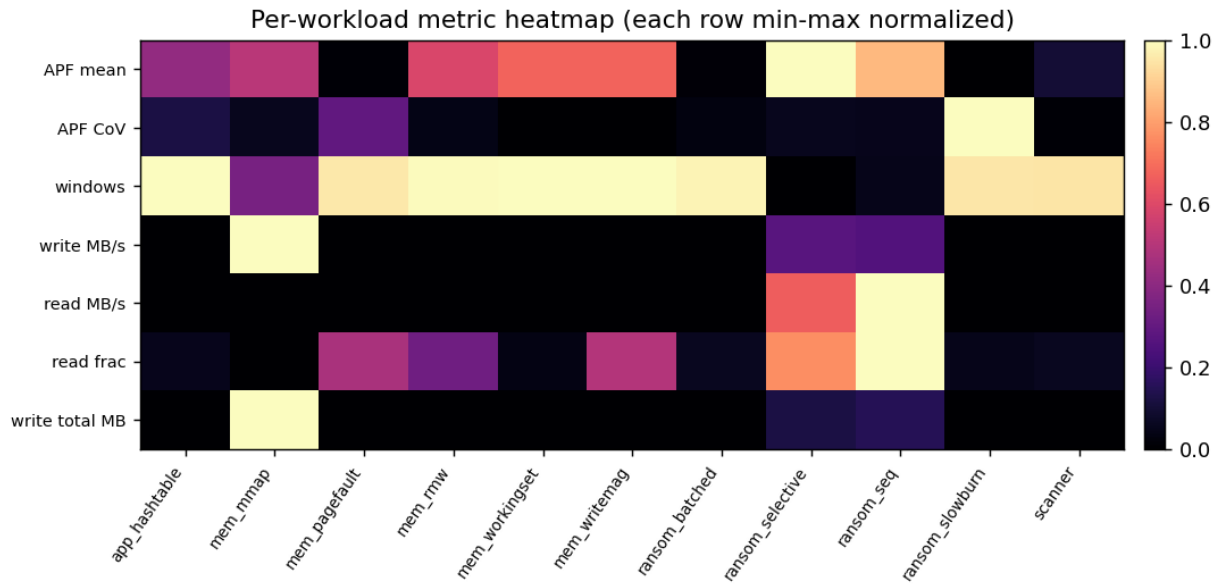
Disk: write rate per workload (log).



Read vs write rate per cell. Encrypt-in-place threats (ransom_seq/selective) read AND write; benign mmap is a pure writer (reads ~0).



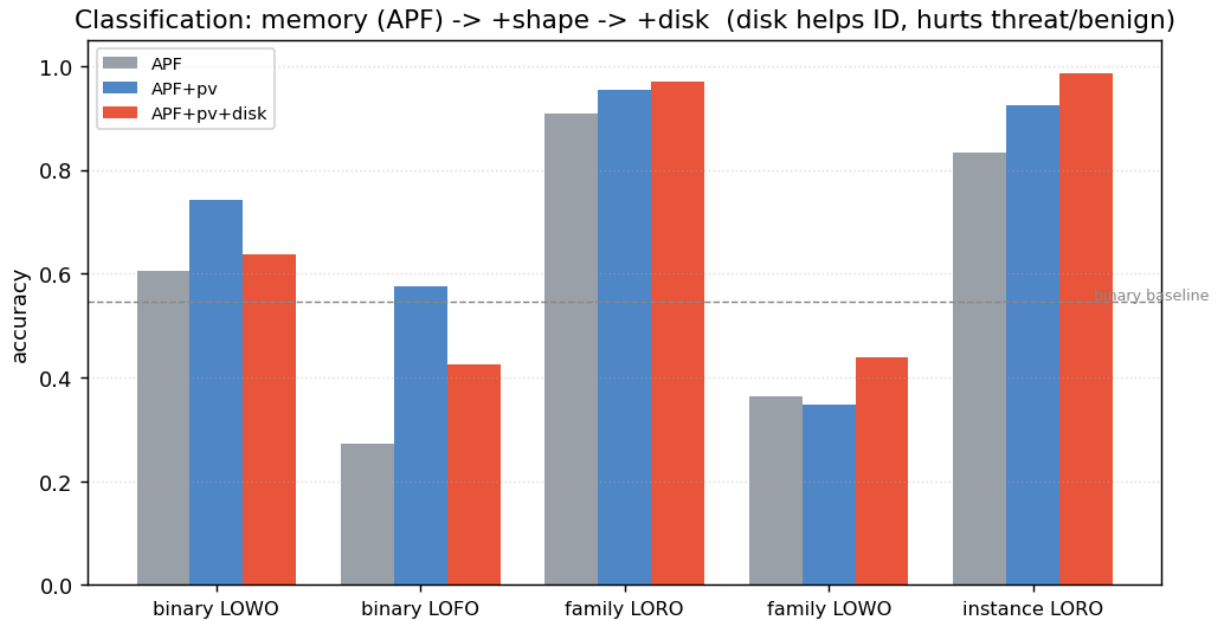
Read fraction of disk I/O per workload. Only the two encryptors read meaningfully; everything else is writes-only.



Per-workload metric heatmap (rows min-max normalized).

family	workload	kind	APF mean	write MB/s	write MB	read MB/s	read frac	win
app	app_hashtable_intensive	benign	0.1550	0.009	9	0.000	0.016	138
mem_fault	mem_mmap_traversal	benign	0.1907	231.636	209083	0.001	0.000	56
mem_fault	mem_pagefault_density	benign	0.0051	0.009	9	0.012	0.148	132
mem_sweep	mem_rmw_intensity	benign	0.2193	0.008	9	0.002	0.104	137
mem_sweep	mem_workingset_sweep	benign	0.2507	0.008	9	0.000	0.012	137
mem_sweep	mem_writemag_sweep	benign	0.2507	0.009	8	0.003	0.155	138
ransomware	ransom_batched	threat	0.0077	0.007	9	0.000	0.019	135
ransomware	ransom_selective	threat	0.3710	62.519	26126	20.749	0.240	12
ransomware	ransom_seq	threat	0.3172	59.070	31500	31.614	0.315	18
ransomware	ransom_slowburn	threat	0.0020	0.293	369	0.004	0.015	131
scanner	scanner_metadata	threat	0.0392	0.269	363	0.005	0.019	131

3. Classification: what each channel buys



Accuracy by feature set across tasks. Disk helps identification, hurts threat/benign detection.

task	APF only	+peak/var	+disk
binary LOWO	0.606	0.742	0.636
binary LOFO	0.273	0.576	0.424
family LORO	0.909	0.955	0.970
family LOWO	0.364	0.348	0.439
instance LORO	0.833	0.924	0.985

Disk I/O (writes + reads) helps characterization — instance 0.924→0.985 (the read signature fingerprints the two encryptors), family LOWO 0.348→0.439 crossing the 0.364 baseline — and hurts detection (binary LOWO 0.742→0.636). Reads sharpen instance ID but do not detect: 3 of 5 threats read ~0, like the benigns. A characterization channel, not a detector.

4. Every cell (66)

fam	workload	kind	dur	rep	APF	win	wMB/s	wMB	rMB/s	rfrac
app	app_hashtable_intensive	benign	120	1	0.1552	48	0.018	10	0.000	0.014
app	app_hashtable_intensive	benign	120	2	0.1570	48	0.013	7	0.000	0.020
app	app_hashtable_intensive	benign	300	1	0.1543	121	0.006	9	0.000	0.017
app	app_hashtable_intensive	benign	300	2	0.1560	121	0.008	11	0.000	0.014
app	app_hashtable_intensive	benign	600	1	0.1540	244	0.003	8	0.000	0.018
app	app_hashtable_intensive	benign	600	2	0.1535	244	0.004	9	0.000	0.015
mem_fault	mem_mmap_traversal	benign	120	1	0.1828	36	277.986	142101	0.001	0.000
mem_fault	mem_mmap_traversal	benign	120	2	0.1760	10	248.990	38693	0.002	0.000
mem_fault	mem_mmap_traversal	benign	300	1	0.1838	89	243.891	333324	0.000	0.000
mem_fault	mem_mmap_traversal	benign	300	2	0.1817	92	283.370	382741	0.000	0.000
mem_fault	mem_mmap_traversal	benign	600	1	0.2022	89	192.433	291447	0.000	0.000
mem_fault	mem_mmap_traversal	benign	600	2	0.2176	21	143.144	66193	0.001	0.000
mem_fault	mem_pagefault_density	benign	120	1	0.0056	47	0.016	8	0.000	0.015
mem_fault	mem_pagefault_density	benign	120	2	0.0058	47	0.017	9	0.072	0.809
mem_fault	mem_pagefault_density	benign	300	1	0.0049	121	0.005	7	0.000	0.019
mem_fault	mem_pagefault_density	benign	300	2	0.0049	121	0.008	10	0.000	0.013
mem_fault	mem_pagefault_density	benign	600	1	0.0046	243	0.003	8	0.000	0.017
mem_fault	mem_pagefault_density	benign	600	2	0.0047	212	0.002	13	0.000	0.015
mem_sweep	mem_rmw_intensity	benign	120	1	0.2190	47	0.015	9	0.000	0.015
mem_sweep	mem_rmw_intensity	benign	120	2	0.2190	47	0.011	9	0.000	0.014
mem_sweep	mem_rmw_intensity	benign	300	1	0.2211	121	0.007	9	0.008	0.547
mem_sweep	mem_rmw_intensity	benign	300	2	0.2183	121	0.008	10	0.000	0.014
mem_sweep	mem_rmw_intensity	benign	600	1	0.2203	243	0.003	8	0.000	0.018
mem_sweep	mem_rmw_intensity	benign	600	2	0.2180	243	0.003	8	0.000	0.017
mem_sweep	mem_workingset_sweep	benign	120	1	0.2510	48	0.015	8	0.000	0.014
mem_sweep	mem_workingset_sweep	benign	120	2	0.2509	47	0.013	8	0.000	0.014
mem_sweep	mem_workingset_sweep	benign	300	1	0.2506	121	0.008	11	0.000	0.009
mem_sweep	mem_workingset_sweep	benign	300	2	0.2507	121	0.006	10	0.000	0.011
mem_sweep	mem_workingset_sweep	benign	600	1	0.2506	244	0.004	11	0.000	0.010
mem_sweep	mem_workingset_sweep	benign	600	2	0.2506	243	0.003	8	0.000	0.014
mem_sweep	mem_writemag_sweep	benign	120	1	0.2509	48	0.018	9	0.000	0.015
mem_sweep	mem_writemag_sweep	benign	120	2	0.2509	48	0.015	7	0.000	0.018
mem_sweep	mem_writemag_sweep	benign	300	1	0.2507	122	0.006	8	0.000	0.017
mem_sweep	mem_writemag_sweep	benign	300	2	0.2507	121	0.008	10	0.000	0.014
mem_sweep	mem_writemag_sweep	benign	600	1	0.2506	244	0.003	8	0.000	0.018
mem_sweep	mem_writemag_sweep	benign	600	2	0.2507	243	0.003	9	0.019	0.850
ransomware	ransom_batched	threat	120	1	0.0077	45	0.013	7	0.000	0.022
ransomware	ransom_batched	threat	120	2	0.0078	47	0.013	7	0.000	0.021
ransomware	ransom_batched	threat	300	1	0.0076	121	0.008	11	0.000	0.014
ransomware	ransom_batched	threat	300	2	0.0076	115	0.004	9	0.000	0.018

fam	workload	kind	dur	rep	APF	win	wMB/s	wMB	rMB/s	rfrac
ransomware	ransom_batched	threat	600	1	0.0077	242	0.004	10	0.000	0.017
ransomware	ransom_batched	threat	600	2	0.0080	240	0.001	8	0.000	0.020
ransomware	ransom_selective	threat	120	1	0.3620	5	69.990	13373	20.686	0.228
ransomware	ransom_selective	threat	120	2	0.3643	7	83.080	15213	24.388	0.236
ransomware	ransom_selective	threat	300	1	0.3580	14	59.624	29766	19.947	0.240
ransomware	ransom_selective	threat	300	2	0.4167	11	68.899	24021	24.806	0.245
ransomware	ransom_selective	threat	600	1	0.3699	23	43.678	46233	16.369	0.244
ransomware	ransom_selective	threat	600	2	0.3548	15	49.841	28152	18.300	0.249
ransomware	ransom_seq	threat	120	1	0.2436	12	79.629	17276	40.194	0.330
ransomware	ransom_seq	threat	120	2	0.3627	9	71.480	18292	37.794	0.313
ransomware	ransom_seq	threat	300	1	0.3503	11	59.579	20552	31.425	0.289
ransomware	ransom_seq	threat	300	2	0.3563	15	63.480	29877	33.785	0.304
ransomware	ransom_seq	threat	600	1	0.3272	26	44.903	43017	24.701	0.327
ransomware	ransom_seq	threat	600	2	0.2632	38	35.348	59985	21.785	0.329
ransomware	ransom_slowburn	threat	120	1	0.0026	45	0.528	260	0.000	0.001
ransomware	ransom_slowburn	threat	120	2	0.0027	46	0.467	260	0.001	0.001
ransomware	ransom_slowburn	threat	300	1	0.0018	115	0.194	328	0.008	0.031
ransomware	ransom_slowburn	threat	300	2	0.0018	115	0.207	326	0.008	0.031
ransomware	ransom_slowburn	threat	600	1	0.0017	234	0.215	601	0.004	0.017
ransomware	ransom_slowburn	threat	600	2	0.0017	233	0.149	439	0.001	0.009
scanner	scanner_metadata	threat	120	1	0.0394	48	0.277	144	0.009	0.029
scanner	scanner_metadata	threat	120	2	0.0392	24	0.340	119	0.005	0.012
scanner	scanner_metadata	threat	300	1	0.0390	122	0.238	308	0.011	0.043
scanner	scanner_metadata	threat	300	2	0.0390	112	0.269	343	0.003	0.012
scanner	scanner_metadata	threat	600	1	0.0394	235	0.257	658	0.002	0.008
scanner	scanner_metadata	threat	600	2	0.0391	244	0.234	605	0.002	0.009